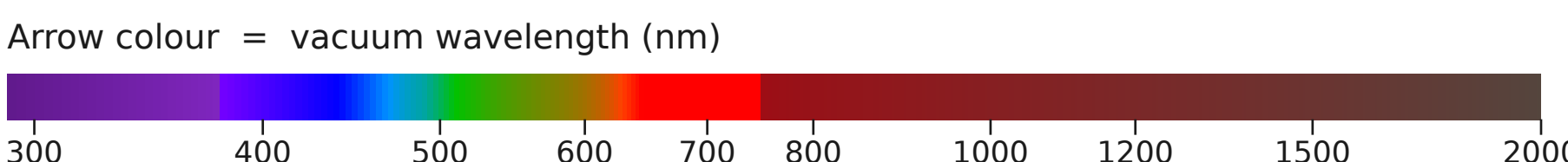


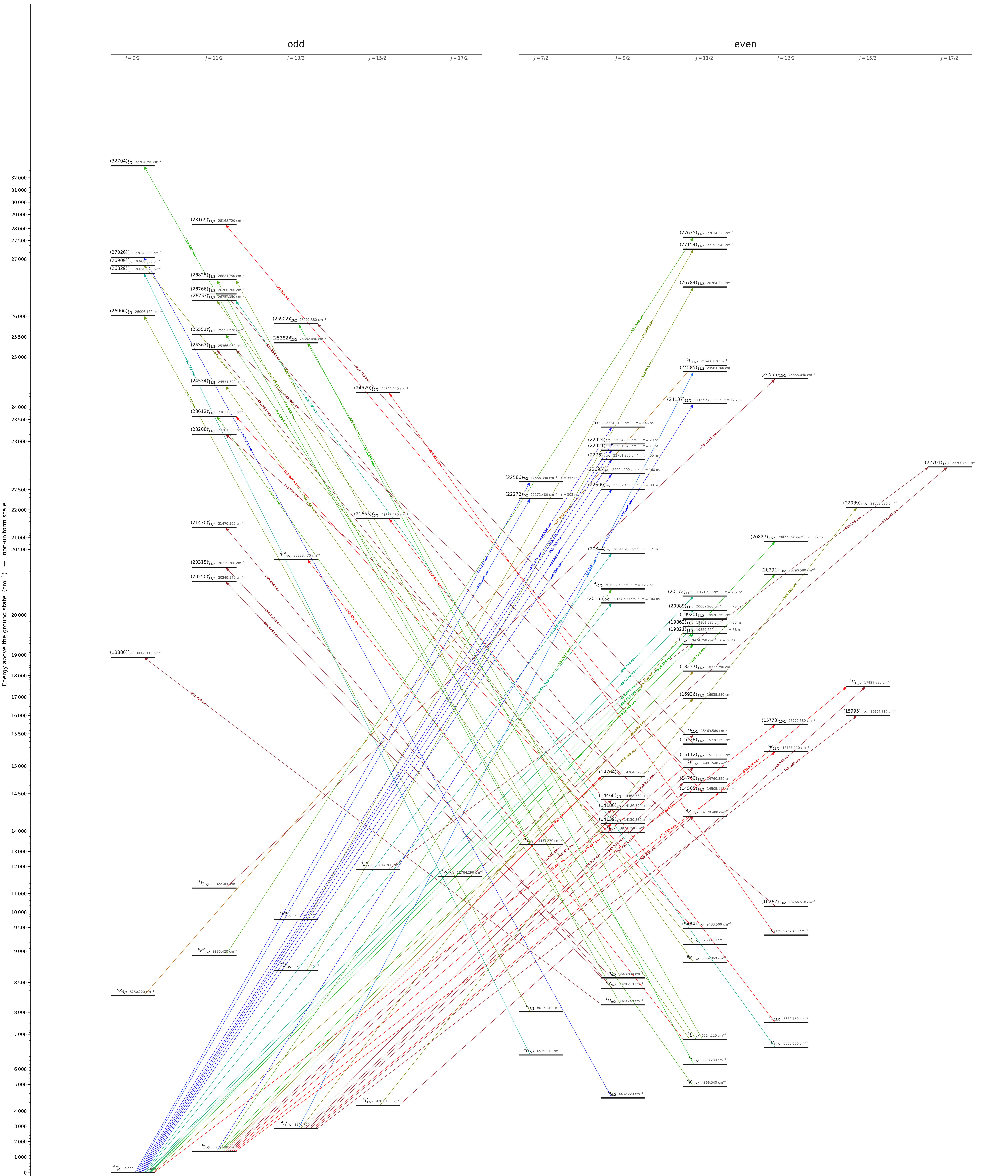
Praseodymium

Arrow label: vacuum wavelength | reduced dipole matrix element |*||*[*l**er*]*l*'|] in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Pr (Pr I). Strongest classified transitions below 2 μm (NIST): 97 levels, 74 lines.

Pr⁺ ionisation limit 44120.00 cm⁻¹ = 5.47018 eV (226.655 nm)



Key transitions									
Transition	λ vac (nm)	λ air (nm)	ν (THz)	[l _{er} l'] (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use	
⁴ I ₀₂ - ⁶ G ₀₂	430.2532	430.1322	696.7815	-	-	1.09	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₀₂ - (22924) ₀₂	436.2166	436.0941	687.2559	-	-	5.488	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₀₂ - (22921) ₀₂	436.2747	436.1521	687.1645	-	-	2.242	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₀₂ - (22762) ₀₂	439.3306	439.2072	682.3846	-	-	2.894	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₁₁₂ - (24137) ₁₁₂	439.3679	439.2445	682.3267	-	-	8.992	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₀₂ - (22695) ₀₂	440.6335	440.5097	680.3670	-	-	0.9474	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₀₂ - (22566) ₀₂	443.1369	443.0125	676.5234	-	-	0.4509	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₀₂ - (22509) ₀₂	444.2588	444.1342	674.8148	-	-	5.305	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₀₂ - (22272) ₀₂	448.9846	448.8586	667.7121	-	-	0.5253	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₀₂ - (20344) ₀₂	491.5387	491.4015	609.9062	-	-	4.681	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₀₂ - (20172) ₁₁₂	495.7428	495.6045	604.7338	-	-	0.686	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₀₂ - (20155) ₀₂	496.1646	496.0262	604.2197	-	-	1.53	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₀₂ - (20089) ₁₁₂	497.7784	497.6396	602.2609	-	-	2.094	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₀₂ - (19862) ₁₁₂	503.4768	503.3364	595.4445	-	-	1.918	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₀₂ - (19821) ₁₁₂	504.5233	504.3827	594.2093	-	-	2.744	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₀₂ - ⁴ I ₁₁₂	513.4854	513.3424	583.8983	-	-	6.121	-	excitation line used for the time-resolved laser-induced...	
⁴ I ₁₁₂ - (20827) ₁₁₂	514.1243	513.9811	583.1128	-	-	2.307	-	excitation line used for the time-resolved laser-induced...	
⁶ I ₁₁₂ - ⁶ I ₀₂	531.5120	531.3642	564.0370	-	-	13.05	-	excitation line used for the time-resolved laser-induced...	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁴¹Pr hyperfine structure (I = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)									
⁴ I ₀₂	926.209	-11.878	F=2: -12740.82	F=3: -9958.63	F=4: -6250.43	F=5: -1617.40	F=6: +3938.96	F=7: +10416.88				
⁴ I ₁₁₂	730.393	-11.877	F=3: -11873.80	F=4: -8948.60	F=5: -5293.56	F=6: -909.64	F=7: +4201.97	F=8: +10039.93				
⁴ I ₁₁₂	613.240	-12.850	F=4: -11503.19	F=5: -8433.04	F=6: -4750.48	F=7: -456.42	F=8: +4448.11	F=9: +9961.94				
⁴ I ₁₁₂	541.575	-14.558	F=5: -11513.77	F=6: -8259.83	F=7: -4465.46	F=8: -131.53	F=9: +4740.97	F=10: +10150.89				
⁴ I ₀₂	923.2(4)	-22(7)	F=2: -12704.08	F=3: -9927.88	F=4: -6228.85	F=5: -1609.18	F=6: +3928.37	F=7: +10380.50				
⁴ K ₁₁₂	867.997	-50.319	F=3: -14125.76	F=4: -10638.41	F=5: -6285.39	F=6: -1070.82	F=7: +3000.36	F=8: +11922.38				
⁴ I ₁₁₂	756.0(10)	-	F=3: -12285.00	F=4: -9261.00	F=5: -5480.00	F=6: -945.00	F=7: +4347.00	F=8: +10395.00				
⁴ H ₀₂	979.0(10)	25(30)	F=1: -11000.36	F=2: -9049.50	F=3: -6120.54	F=4: -2210.25	F=5: +2685.64	F=6: +8572.50				
⁴ K ₁₁₂	755.456	-48.633	F=4: -14183.50	F=5: -10391.26	F=6: -5846.74	F=7: -553.31	F=8: +5485.10	F=9: +12264.00				
⁴ L ₁₁₂	474.692	-29.633	F=3: -7726.00	F=4: -5818.18	F=5: -3437.05	F=6: -585.01	F=7: +2735.00	F=8: +6519.61				
⁴ L ₁₁₂	776.286	-43.592	F=4: -14572.13	F=5: -10677.29	F=6: -6009.01	F=7: -570.31	F=8: +5635.28	F=9: +12603.75				
⁴ I ₀₂	168.0(10)	-	F=1: -1890.00	F=2: -1554.00	F=3: -1050.00	F=4: -378.00	F=5: +462.00	F=6: +1470.00				
⁴ H ₀₂	797.0(20)	-	F=2: -10958.75	F=3: -8567.75	F=4: -5379.75	F=5: -1394.75	F=6: +3387.25	F=7: +8966.25				
⁴ K ₀₂	213.531	-4.136	F=2: -2937.95	F=3: -2296.11	F=4: -1440.82	F=5: -372.47	F=6: +908.40	F=7: +2401.19				
⁴ K ₀₂	255.0(20)	-	F=2: -3506.25	F=3: -2741.25	F=4: -1721.25	F=5: -446.25	F=6: +1083.75	F=7: +2868.75				
⁴ I ₀₂	797.0(20)	-	F=2: -10958.75	F=3: -8567.75	F=4: -5379.75	F=5: -1394.75	F=6: +3387.25	F=7: +8966.25				
⁴ I ₁₁₂	854.297	-31.807	F=4: -16030.30	F=5: -11749.03	F=6: -6615.54	F=7: -632.04	F=8: +6198.91	F=9: +13874.37				
⁴ K ₁₁₂	769.0(10)	-30(20)	F=3: -12508.66	F=4: -9423.50	F=5: -5570.72	F=6: -952.79	F=7: +4427.34	F=8: +10566.25				
⁴ K ₁₁₂	949.091	-13.721	F=3: -15428.40	F=4: -11627.85	F=5: -6878.84	F=6: -1182.50	F=7: +5459.83	F=8: +13046.57				
⁴ L ₁₁₂	977.0(10)	-24(20)	F=3: -15886.18	F=4: -11970.85	F=5: -7079.63	F=6: -1214.49	F=7: +5622.22	F=8: +13427.75				
⁴ K ₁₁₂	1056.0(10)	-15(10)	F=4: -19805.77	F=5: -14521.15	F=6: -8181.52	F=7: -787.90	F=8: +7658.48	F=9: +17156.25				
(9484) ₁₁₂	731.0(10)	-15(10)	F=3: -11884.95	F=4: -8956.37	F=5: -5297.49	F=6: -909.52	F=7: +4206.05	F=8: +10047.50				
⁴ K ₁₁₂	991.907	-7.246	F=4: -18601.04	F=5: -13639.28	F=6: -7686.08	F=7: -741.95	F=8: +7192.52	F=9: +16116.68				
(10267) ₁₁₂	972.0(20)	-	F=4: -18225.00	F=5: -13365.00	F=6: -7533.00	F=7: -729.00	F=8: +7047.00	F=9: +15795.00				
⁴ I ₁₁₂	1272.0(10)	75(50)	F=3: -20638.98	F=4: -15573.89	F=5: -9233.32	F=6: -1611.14	F=7: +7300.02	F=8: +17508.75				
⁴ L ₁₁₂	355.0(20)	-	F=5: -7543.75	F=6: -5413.75	F=7: -2928.75	F=8: -88.75	F=9: +3106.25	F=10: +6656.25				
⁴ I ₀₂	854.0(10)	-15(15)	F=2: -11749.38	F=3: -9182.88	F=4: -5762.62	F=5: -1490.12	F=6: +3632.75	F=7: +9603.75				
⁴ K ₁₁₂	986.0(10)	-20(10)	F=3: -16030.77	F=4: -12060.66	F=5: -7145.48	F=6: -1226.86	F=7: +5673.23	F=8: +13552.50				
(14186) ₀₂	910.0(10)	-60(30)	F=2: -12540.00	F=3: -9792.00	F=4: -6135.00	F=5: -1575.00	F=6: +3880.50	F=7: +10222.50				
(14468) ₀₂	762.0(20)	20(20)	F=2: -10468.33	F=3: -8188.33	F=4: -5146.00	F=5: -1339.33	F=6: +3234.17	F=7: +8577.50				
(14505) ₁₁₂	777(3)	-	F=3: -12626.25	F=4: -9518.25	F=5: -5633.25	F=6: -971.25	F=7: +4467.75	F=8: +10683.75				
(14764) ₀₂	867.0(20)	-	F=2: -11921.25	F=3: -9320.25	F=4: -5852.25	F=5: -1517.25	F=6: +3684.75	F=7: +9753.75				

Measured A, B constants; sources in the data file.